

Smart Navigation Strategies for Autonomous Water Surface Cleaning Rover

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1. Introduction:

Pollution of water due to floating debris creates significant environmental problems, affecting both aquatic ecosystems and human well-being. Conventional techniques for cleaning water surfaces, such as manual work and typical mechanized systems, tend to be costly and ineffective. This study aims to improve autonomous robotic systems for cleaning water surfaces, tackling existing challenges in navigation, obstacle evasion, and coverage path planning (CPP). The suggested system will enhance environmental perception by incorporating advanced CPP techniques, artificial intelligence (AI), and real-time obstacle avoidance while employing a mix of sensors such as RGB cameras, LiDAR, and an inertial measurement unit (IMU). A Nonlinear Model Predictive Control (NMPC) framework will enable real-time trajectory modifications, paired with an adaptive path planning algorithm to enhance coverage and energy efficiency. The system will be validated via simulations and practical tests, aiming for improved decision-making and navigation for effective water pollution management.

2. Problem Statement:

Current autonomous water cleaning robots face several critical challenges that reduce their effectiveness:

- **Inefficient Path Planning:** Many existing robots rely on predefined static routes, leading to incomplete coverage and redundant cleaning loops.
- **Poor Obstacle Avoidance:** Obstacle detection mechanisms often fail to account for dynamic floating debris and moving obstacles, resulting in collisions or ineffective rerouting.
- **Limited Environmental Adaptability:** Current models struggle to adapt to changing water currents, unpredictable waste distribution, and varying environmental conditions, reducing overall efficiency.

3. Objectives:

The primary objectives of this research are:

- To create a robust water surface cleaning robot system architecture with seamless addition of sensing, optimal navigation and the control mechanisms.
- To use enhanced nonlinear model predictive control (NMPC) framework for obstacle avoidance and enhance the adaptability.
- To put into practice a path planning algorithm that is optimized for full water coverage.
- To include real-time environmental sensing to dynamically tune the navigation and cleaning operations in response to the dynamic water patterns.
- To validate the proposed or suggested mechanical system - a combination of real life experiments and precision driven physics based simulations will be one of the primary objectives.

4. Methodology:

4.1 System Design:

The suggested autonomous water surface cleaning robot will have a twin-pontoon hull design that improves stability and agility in dynamic aquatic situations. A front-mounted garbage collecting module allows for more effective waste gathering and debris removal. The STM32 microcontroller will be the primary processing unit, handling real-time navigation, sensor fusion, and motor control. To accomplish precise localization and environmental awareness, the STM32 will communicate with a high-precision RTK-GNSS module, an inertial measurement unit (IMU) for motion tracking, a 3D LiDAR sensor for better obstacle recognition, and an RGB camera for visual data processing. 3D LiDAR is chosen over millimeter-wave radar

because of its better resolution and improved navigation capabilities. The microprocessor will continually evaluate incoming sensor data to provide optimal navigation commands, resulting in smooth trajectory corrections and accurate maneuverability.

The propulsion system will use T200 Blue Robotics thrusters, which provide high-efficiency propulsion with improved movement control. The STM32 microprocessor will electronically operate these thrusters, allowing the robot to move forward, backward, and spin on its axis to provide ideal cleaning coverage. A watertight electronic compartment (IP67 certified) protects internal electronics from water exposure, assuring system endurance and dependability. In addition, an automated garbage trapping module will modify its opening and shutting mechanism based on the robot's moving condition in order to properly keep gathered rubbish. To improve power efficiency and decrease heat generation in motor drivers, the system will be supplied by a 48V lithium battery, allowing for longer working duration while minimizing energy usage. The suggested design intends to increase the efficiency, flexibility, and dependability of water surface cleaning robots in complicated aquatic settings by including sophisticated navigation sensors, a better propulsion system, and an upgraded power supply.

4.2 Navigation and Obstacle Avoidance

Navigation and obstacle avoidance will be achieved through a combination of sensor fusion and advanced control algorithms, all managed by the STM32 microcontroller. The STM32 will process real-time data from the GNSS and IMU sensors to determine the robot's precise position and orientation. The GNSS will provide global positioning data, while the IMU will assist in motion tracking, allowing the robot to maintain stability in varying water conditions.

To detect obstacles, the millimeter-wave radar and RGB camera will collaborate to recognize floating debris, immobile items, and moving obstacles like boats. The STM32 microcontroller will combine these sensor readings to develop a complete environmental model, enabling the robot to make educated decisions regarding route changes. A refined Nonlinear Model Predictive Control (NMPC) algorithm will be utilized to boost the robot's capability to follow its intended route while adaptively responding to identified obstacles.

For effective obstacle avoidance, the STM32 microcontroller will employ a real-time path replanning approach. Upon detecting an obstacle, the microcontroller will perform a local path adjustment, allowing the robot to maneuver around the obstacle while staying close to the original coverage route. A feedforward control system will be implemented to counteract disturbances like water currents and changes in hull weight caused by collected debris. By utilizing the fast processing abilities and effective power management of STM32, the system will provide responsive and accurate navigation, guaranteeing optimal coverage efficiency and safe functioning in various aquatic environments.

4.3 Path Planning and Simulation Techniques

The path planning will be developed using a combination of A* (A-star) and Dijkstra's algorithm which are already well established for this purpose. Dijkstra's algorithm finds the shortest path between two nodes while A* has a smarter and a more analytical approach that allows it to navigate obstacles. Hence combining the efficiency of Dijkstra's algorithm with the complexity of A* will provide us with one of the most efficient path finding methods. However, we will take it a step further by also integrating real-time-decision-making capabilities in the system, making it also capable of adapting to the environment dynamically. This will be especially helpful due to the complexity of real world situations - factors like water currents, varying trash densities, obstacles, or even the weather's effects - require case by case responses. By using the onboard sensors (mentioned previously), the system will continue to update its path planning accordingly for the optimal flow.

Before real-world deployment, extensive testing will be conducted through simulations in environments like MATLAB or ROS (Robot Operating System). MATLAB will be used for assessing the system in various different simulation cases. This will allow us to validate the system by observing and optimizing its responses throughout the various different factors in simulation, like water current, obstacle maneuvering and pathfinding. ROS simulations will further help in testing the interaction between the robot and its sensors, as well as verifying the communication between software components, sensor data processing, and the control system. We plan on using the virtual environment function in ROS to make more realistic simulations for the several dynamic factors in the real world like changes in environment. The combination of the data from both MATLAB and ROS

will determine what we should expect from our cleaning robot in real life scenarios. This means we can somewhat predict the issues we may face, as well as some counter measures we may need to develop to combat these issues.

Finally, we will move on to real-world testing. We will conduct the testing on controlled water bodies at first, meaning we can change various variables mentioned earlier, like the water current, the trash location and densities, obstacle navigation, etc. Afterwards we plan on moving on to actual field testing in natural water bodies where the system will be put into the ultimate test. Key performance indicators (KPIs) such as waste collection rate, battery consumption, time taken to cover specific areas, and obstacle avoidance success will be measured during these field tests. The finalized data that we will obtain at the end of all these testing should provide us with the necessary information for the maximum optimization of our system so that it may perform as efficiently as possible even during large scale deployments.

5. Expected Outcomes

The project aims to deliver a fully functional water cleaning bot with an advanced navigation system that will allow it to efficiently detect and collect wastes in water bodies, all the while avoiding obstacles and handling natural dynamic conditions associated with those environments. Another aim is the implementation of an optimized navigation algorithm that will provide the most efficient path planning and navigation as well as adaptability to more complex scenarios. The system's performance will be validated through both simulations and real-world testing, confirming its efficiency and reliability in various environments. Ultimately, this project will contribute to advancements in autonomous environmental cleaning technologies.

6. Timeline

Phase	Duration
Literature Review & Problem Analysis	1 month
System Design & Hardware Selection	2 months
Sensor Integration & Navigation Development	3 months
Path Planning & AI Implementation	2 months
Simulation & Testing	2 months
Final Evaluation & Report Writing	1 month

7. References

Zhu, J., Yang, Y., & Cheng, Y. (2022). SMURF: A Fully Autonomous Water Surface Cleaning Robot with A Novel Coverage Path Planning Method. *Journal of Marine Science and Engineering*, 10(11), 1620. <https://doi.org/10.3390/jmse10111620>

Request for Approval

We kindly seek your approval and guidance for this research proposal. Your feedback and insights would be invaluable in refining the study and ensuring its success. Looking forward to your response.